COSMOLOGY

Problem Set 1: The Problems In Cosmology

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These are the end of chapter problems for Chapter 1 of the notes. Equation numbers refer to the full notes.

1. Newton's Shell Theorem

Newton's Shell Theorem states that

(1) a spherically symmetric body affects external objects gravitationally as though all of its mass were concentrated at a point at its center, and
(2) if the body is a spherically symmetric shell (i.e., a hollow ball), no net gravitational force is exerted by the shell on any object inside, regardless of the object's location within the shell.

a). Let us first start by using the universal law of gravitation. For a continuous mass distribution, the force on a test mass m_s at position $\boldsymbol{r'}$ is

$$\boldsymbol{F}_g = -Gm_s \int_V \rho(\boldsymbol{r}) \frac{\boldsymbol{r'} - \boldsymbol{r}}{|\boldsymbol{r'} - \boldsymbol{r}|^3} \, d^3r$$

where $\rho(\boldsymbol{r})$ is the mass density and V is the volume of the mass that we are integrating over. Assuming that the mass density of a thin spherical shell is uniform, show that the force inside the shell is 0 for every $\boldsymbol{r'}$ inside the sphere.

b). Suppose that the density is not constant but still only depends on the radius, $\rho(r)$. Does this result still hold true?

c). Now take $\boldsymbol{r'}$ to be outside the sphere. Show that the gravitational force reduces down to that of 2 point masses separated by some distance d, where d is the distance between the centers of mass of the 2 objects.

d). Using Gauss's law for gravity

$$\oint_S \boldsymbol{g} \cdot d\boldsymbol{A} = -4\pi G M_{\text{enc}}$$

where $\boldsymbol{g}$ is the gravitational field and M_{enc} is the mass enclosed by the surface S, show that you retrieve the exact same results with almost no work.

2. Dark Matter In Galaxies

In the dark matter section, we argued that if the mass distribution dies off at large distances from the galactic center, then the dominating variable is the $1/r$ in (1.3). Now consider a more realistic distribution of the form

$$\rho(r) = \frac{\rho_0}{(1 + r/r_0)^\alpha}$$

a). Derive the rotation curve $v(r)$ for a value of α which gives a flat rotation curve at $r \gg r_0$ as indicated by measurements.

b). At $r = 10^5$ lightyears the measurement yields $v_{\text{meas}} = 225$ km/s, while the visible matter alone predicts $v_{\text{calc}} = 15$ km/s. Calculate the visible as well as the true galaxy mass enclosed within that radius. What is the percentage of dark matter in the galaxy? What is the average dark matter mass density? (Use $G = 6.67 \times 10^{-11}$ m^3 kg^{-1} s^{-2}.)

3. Hubble's Law

a). Show that if the universe expands uniformly, so that every distance is multiplied by the same factor $a(t)$, then every observer sees the other galaxies receding with a speed proportional to their distance, $v = Hd$ with $H = \dot{a}/a$. Why does this mean that being at the "center" of the expansion is meaningless?

b). With $H_0 = 70$ km s^{-1} Mpc^{-1}, compute the **Hubble time** $1/H_0$ in years and the **Hubble distance** c/H_0 in Mpc and in lightyears. These set the rough age and size of the observable universe.